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Experiment 3:

Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree

uploaded = files.upload()
filename = list(uploaded.keys())[0]
data = pd.read_csv(filename)
encoders = {}

for column in data.columns:
    le = LabelEncoder()
    data[column] = le.fit_transform(data[column])
    encoders[column] = le

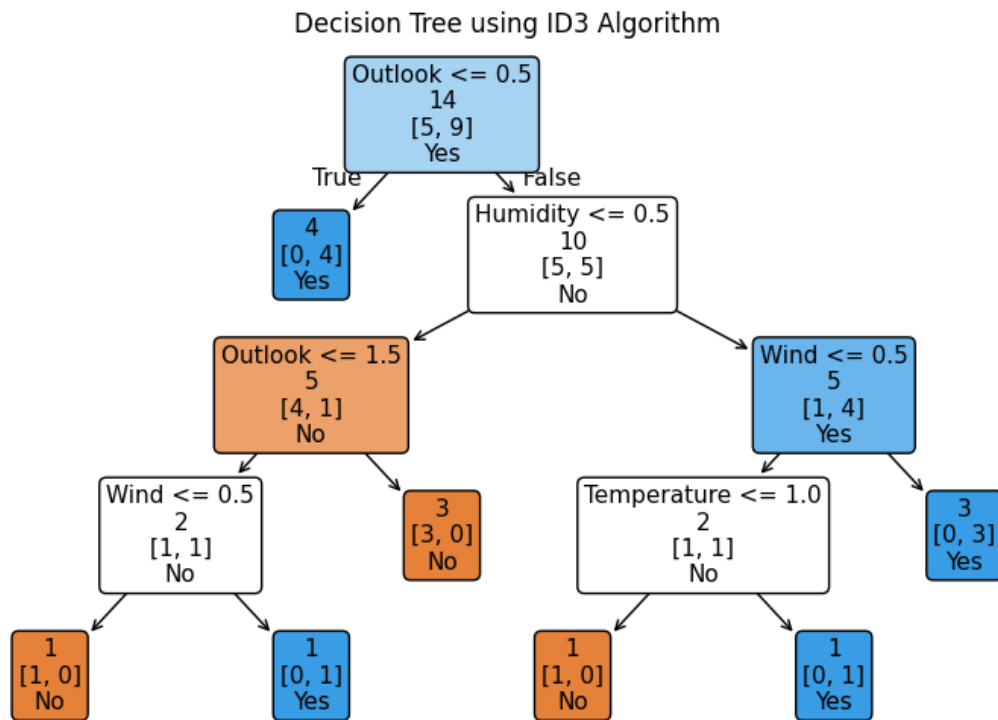
X = data.drop("PlayTennis", axis=1)
y = data["PlayTennis"]
model = DecisionTreeClassifier(
    criterion="entropy",
    random_state=42
)
model.fit(X, y)
plt.figure(figsize=(10, 6))
plot_tree(
    model,
    feature_names=X.columns,
```

```

class_names=encoders["PlayTennis"].classes_,
filled=True,
rounded=True,
impurity=False,
label="none",
fontsize=11
)
plt.title("Decision Tree using ID3 Algorithm")
plt.show()
new_sample = pd.DataFrame({
    "Outlook": ["Sunny"],
    "Temperature": ["Cool"],
    "Humidity": ["High"],
    "Wind": ["Weak"]
})
for column in new_sample.columns:
    new_sample[column] = encoders[column].transform(new_sample[column])
prediction = model.predict(new_sample)
result = encoders["PlayTennis"].inverse_transform(prediction)
print("\nNew Sample: Sunny, Cool, High, Weak")
print("Prediction:", result[0])

```

Output:



New Sample: Sunny, Cool, High, Weak

Prediction: No